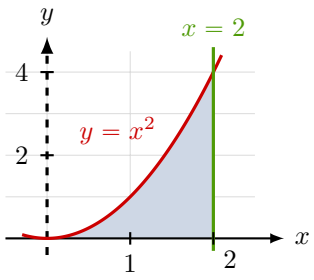
L4.3b Which Method?

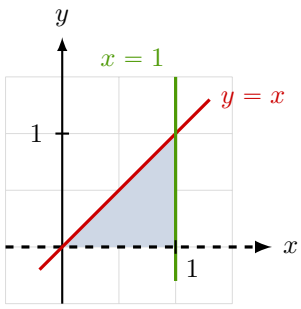
Recall: volumes by cylindrical shells

Name _____ **Solo – 9 minutes**

1. The region bounded by $y = x^2$, $y = 0$ and $x = 2$ is rotated about the y-axis. Draw a representative shell, label r and $h(r)$, and use shells to find the volume.



2. The region bounded by $y = x$, $y = 0$ and $x = 1$ is rotated about the x-axis. Draw a representative shell, label r and $h(r)$, and write the volume as an integral in r. Do not evaluate.



3. For each, is the representative rectangle perpendicular or parallel to the spin axis?

(a) a disk (b) a washer (c) a shell